

Technical Document #943: Database Systems - Comprehensive Overview

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Database replication copies data across multiple servers. It improves availability and provides fault tolerance.

Database monitoring tracks performance metrics and identifies bottlenecks. Query analysis and indexing strategies optimize database performance.

Database indexing improves query performance by creating data structures that allow faster data retrieval.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Key Takeaways:

- Sharding distributes data across multiple database servers.
- Database connection pooling reuses database connections to reduce overhead.
- NoSQL databases handle unstructured and semi-structured data.